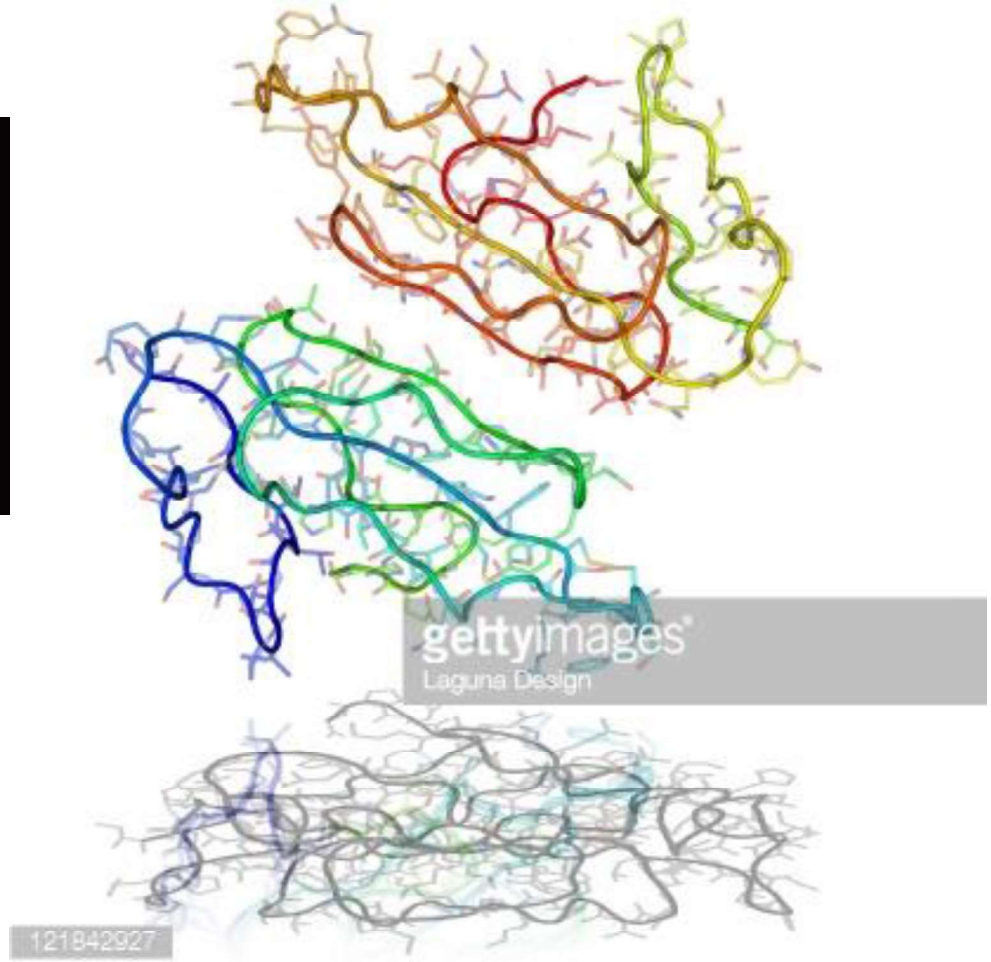
Venom Toxins

- **Neurotoxin:** inhibit neurotransmissions by blocking the exchange of positive and negative ions across the neuronal membrane lead to no nerve impulse, thereby paralysing the nerves.
- **Cardiotoxins:** are components that are specifically toxic to the heart.
- **Hemotoxin:** cause hemolysis, the destruction of red blood cells (erythrocytes), or induce blood coagulation (clotting).
- **Cytotoxins:** are components that kill cell and cellular mechanism.

Neurotoxin



Bungarotoxins

- **Bungarotoxins** are a group of closely related neurotoxic proteins derived from the venom of kraits
- There are three types:
- α-Bungarotoxin
- β-Bungarotoxin
- γ-Bungarotoxin

https://www.youtube.com/watch?v=nb-rnL3z1Lc

https://www.youtube.com/watch?v=lCPRlcqhfhU

Neurotoxin

- **Chlorotoxin** is a 36-amino acid peptide found in the venom of the deathstalker scorpion

- AA Sequence

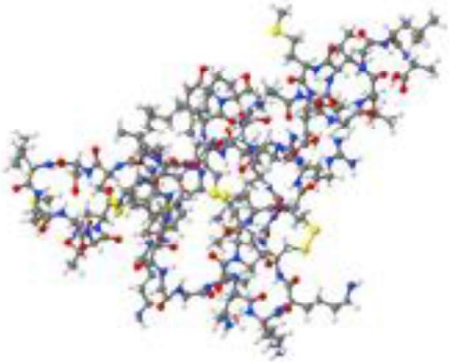
  MCMPCFTTDHQMARKCDDCCGGKGRGKCYGPQCLCR

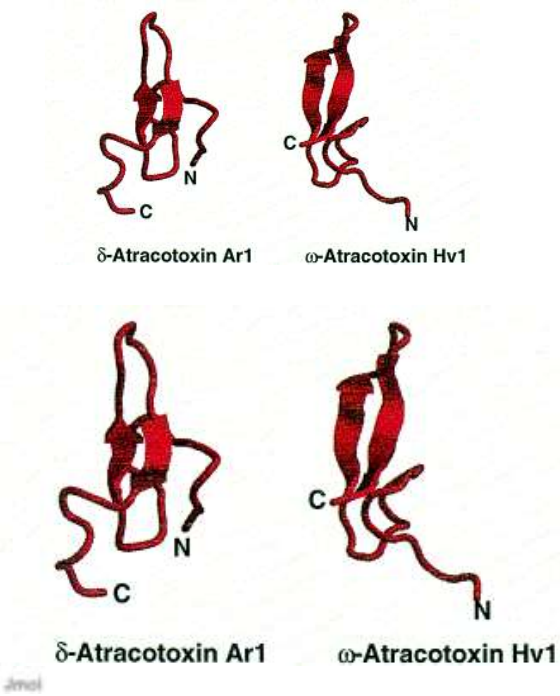
Apitoxin

Apitoxin, or honey bee venom, is a bitter colourless liquid; its active portion a mixture of proteins, which causes local inflammation and acts as an anticoagulant.

Spider toxins

- **Spider toxins** are a family of proteins produced by spiders which function as neurotoxins. The mechanism of many spider toxins is through blockage of calcium channels.





Use of snake venoms to treat disease

- Snake venom contains many biologically ingredients, some may be useful to treat disease.

1. Phospholipases type A2 (PLA2s) from the Tunisian vipers has been found to have **anti-cancer** activity.

2. Phospholipases A2 hydrolyze phospholipids and thus could act on bacterial cell surfaces, providing novel **antimicrobial** (antibiotic) activities.

3. The **analgesic** (pain-killing) activity of many snake venom proteins has been long known but problem is delivery.